Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/10/2010
PI Name:	Cornforth, Michael Ph.D.		
Project Title:	Structural Chromosome Aberrations Formed in Response to Changes in Proton Energy and Dose Rate		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mcornfor@utmb.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-772-4244
Organization Name:	University of Texas Medical Branch		
PI Address 1:	301 University Blvd		
PI Address 2:	Radiation Oncology		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: NCE until 8/31/2012, per C. Guidry/JSC (Ed., 9/26/2011) NOTE: Received NCE until 8/31/2011 (original end date was 8/31/2010), per C. Guidry/JSC (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Loucas, Bradford (University of Texas Medical Branch)		
Grant/Contract No.:	NNX07AP83G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	From the perspective of NASA, risk uncertainties related to radiation exposure during space exploration are unacceptably large. These include the effects of sporadic solar particle events (SPEs) which can produce relatively high levels of radiation, principally from protons. The resultant dose to personnel from an SPE can be high enough to warrant concerns of cancer induction. Because human epidemiological data in this area are not available, reduction in risk uncertainties will require data from meaningful surrogate endpoints, such as chromosome damage. A reasonable and widely held notion is that dose rate effects from protons will be the same as for x- and gamma rays. There is, however, not much direct experimental evidence to support this contention, which adds additional uncertainty to risk projections. We propose to simulate an SPE using proton emissions from the National Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). Owing to differences in ionization density that may translate to differences in biological effect, proton energies of 50 MeV and 250 MeV will be studied. These will be delivered at both high and low dose rates. Results will then be compared to those obtained for gamma rays from a 137-Cesium source. Much is known about chromosomal response to other types of radiation, such as x- and gamma rays, including how the level of damage is influenced by the rate at which radiation is delivered. We have chosen structural damage to chromosomes of human cells in culture as a surrogate endpoint for cell death and cancer. We first propose to construct quantitative dose responses for the induction of chromosome aberrations found in cells when they arrive at the first mitosis following exposure. For this and other objectives, multiplex fluorescence in situ hybridization (mFISH) will be used, which effectively paints each of the 23 pairs of human chromosomes a unique color. This allows even very complicated rearrangements among all chromosomes to be visualized simultaneously. The second set of objectives involves analysis of chromosome damage in clonal descendents of single cells that survive exposure to radiation. Several such clones will be expanded in culture and examined by mFISH for transmissible chromosome aberrations. A subset of these clones will be further examined for structural aberrations that are too small to be seen under the microscope, by employing comparative genomic hybridization applied to DNA microarrays (aCGH). It is argued that genetic damage in cells surviving radiation may be diagnostic of the type of radiation encountered, and that such damage has more direct relevance to carcinogenesis.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The overarching objective of these studies is to help reduce uncertainties associated with exposure to ionizing radiation in a space environment, specially those associated with exposure to solar particle events (SPEs) that relate chiefly to high energy proton exposure. Currently the assumption is that such protons behave essentially like other forms of low LET (linear energy transfer) photons such as X- and gamma rays, for which much quantitative information already exists. The extent to which the clastogenic (chromosome aberration producing) effects of protons actually are (or are not) similar to photons in the context of energy and dose rate is an important piece of information needed to help reduce the aforementioned uncertainties.		

Task Progress:	On the basis of biophysical modeling described in the last progress report, and within the constraints of NSRL capabilities in terms of scheduling and ion availability, we determined energies and dose rates that were compatible with our scientific objectives. For the spring run these were 150 MeV protons and 1 GeV protons, both of which were delivered at high (30 cGy/min) and low (30 cGy/h) dose rates to telomerase-immortalized human fibroblasts to total absorbed doses of 0, 100, 200, and 300 cGy. In the fall run 50 MeV protons became available, and this energy was used instead of the previous 150 MeV treatment group. As in the spring, 1 GeV protons were used again, which served as a repeat. Progress as regards the three objectives listed in the original proposal are discussed below. Objective 1. Cells will be scored for structural aberrations in chromosomes at the first mitosis following exposure by 24-color multiplex fluorescence in situ hybridization (mFISH). This represents perhaps the major objective of the proposal, insofar as NASA is concerned. Not counting the zero-dose sham-irradiated control, chromosome preparations were made for all twelve treatment groups for both the spring and fall runs. Each such treatment constituted at least two postirradiation collection intervals, bringing the total to =48 samples. In our last progress report, a total of 91 cells had been analyzed. Since then we have analyzed a total 678 cells by mFISH. We assembled a “completion matrix” to track progress made for this objective, which refers to data that must eventually be tallied for all 24 treatment groups. Out of the 24 treatment groups, there are eight instances where no images have yet been scored by mFISH, and another four instances where more images are required. In addition, it should be noted that in the original grant application, we proposed comparing results from protons to those obtained using 137Cs gamma rays delivered at 0.5 cGy/min using the low dose rate facility Colorado State University. Although the data set is not yet complete, we were able to construct dose response relationships for three different energies of protons (50 MeV, 150 MeV and I GeV), delivered at both high and low dose rates. Including zero-dose controls, four doses were given: 0.0, 1.0, 2.0 and 3.0 Gy. A significant point of interest is the high degree of curvature seen in the dose response observed for high energy (1 GeV) protons for the spring run, due mainly to the single dose point at 3 Gy. This result runs counter to our expectations, as well as at least one other NASA-funded cytogenetic study (J.S. Bedford, PI) which showed little if any difference in the yield of aberrations as a function of proton energies in the range of 50 MeV to 1 GeV. The fall run contained a repeat of this treatment group, in which the dose response following 1 GeV protons was similar to that observed for 150 MeV particles. When finished, there will be six data points associated with each absorbed dose, but as it stands, this is presently only possible for the 3 Gy dose. Nevertheless, we can conclude thus far that the frequencies of cytogenetic damage is not strongly dependent on proton energy, although there exists a trend for higher energies to produce more damage per unit dose. Furthermore, there exists a strong dose rate effect between samples irradiated at low versus high radiation intensities. The extent to which this effect is quantitatively similar to gamma photons will require a direct comparison to 137Cs gamma rays at both dose rates, which will require further experiments conducted at Colorado State University. Objective 2. Isolation and mFISH analysis of independent cell clones surviving proton irradiation. In the original proposal, cultures were to be given either 2 or 4 Gy (from Objective 1) and cryogenically preserved, later to be thawed and replated at low density for clonal expansion. We proposed to select a representative number of independent clones surviving high dose rate exposure to 4 Gy of 50 MeV protons. These were to be expanded in culture, and assessed for transmissible chromosome rearrangements. Owing to practical constraints placed upon us at NSRL during the actual irradiations during the spring run, 3-Gy doses of either 150 MeV or 1 GeV were used instead. But in addition to high dose rate (HDR) exposures initially planned, low dose rates (LDR) groups were also examined. A total of 93 clones were isolated from the following treatment groups: HDR and LDR for 150 MeV protons, and HDR and LDR for 1 GeV protons. We found that we were able to begin clonal isolation immediately following the shipment of cultures back from Brookhaven, which obviated the need to freeze cultures down upon arrival. The clonal derivatives were cryogenically preserved for subsequent mFISH analysis and for possible use in Objective 3. Since the previous report, these 93 clones have now been analyzed for the spring run by 24-color mFISH. The pooled data for the four treatment groups show that 22/93, or about 23 percent, of the recovered clones contained rearrangements. No recurrent aberrations were observed, each clone harboring its own unique constellation of rearrangements, which were confined to translocations. No instances of dicentric or ring chromosomes were observed. About twenty mFISH metaphase karyotypes were analyzed for each clone shown. It is significant that each of the cells within individual aberration-bearing clones were homogenous for their respective unique rearrangements, and therefore that no evidence for ongoing chromosomal instability was found. The relatively small number of aberration-bearing clones in each treatment group do not allow quantitative inferences to be made regarding energy or dose rate. Only two of the 22 aberrant clones contained stable complex exchanges. Curiously, both derived from 1 GeV protons delivered at the low dose rate. Objective 3. CGH microarray analysis for the detection of submicroscopic deletions and co-deletions associated with transmissible exchange aberrations. In the original proposal we had planned to extract DNA from a subset of isolated clones, which was then to be differentially fluor-labeled and hybridized to a CGH microarray (aCGH) to detect transmissible deletions, possible amplifications, and for evidence of submicroscopic co-deletion at the breakpoints of chromosome translocations. We made substantial preliminary progress on this aim since the last reporting period. To date our analyses are limited to a total of 10 clones derived from HDR 137Cs gamma irradiation and two clones derived from 1 GeV 56Fe ion treatment. In the last progress report, we were in the process of shipping DNA samples to NimbleGen for high-density aCGH. Since then, the samples have been analyzed and processed. They included DNA isolated from one unirradiated master control culture (K-1 mother clone) and DNA from independent clones surviving exposure to the following treatments: a. Five clones exposed to 4Gy of 137Cs gamma rays, each of which was homogeneous for one or more chromosome translocations. b. Four such gamma-irradiated clones the were cytogenetically normal by mFISH c. Two clones recovered from cultures exposed to 1 Gy of 1 GeV/amu 56Fe ions, each of which homogeneous for one or more translocations. The log2 ratios of red:green fluorescence for all 24 chromosome types, comparing control DNA with that derived from clones surviving radiation exposure, were produced so that deletions could be detected. We have not yet analyzed by aCGH enough clones surviving 56Fe ion irradiation to make conclusions regarding the size-frequency distributions of deletions. However, preliminary pooled aCGH data for the ten clones surviving 137Cs gamma rays show some interesting results. Frequency histograms, when binned in large 0.5 Mb intervals show that, while smaller sized deletions are recovered in higher frequencies, there exists an extended tail to the distribution, and that some deletions recovered span several megabases. When such a histogram is binned at 10 kb intervals, in order to highlight smaller deletions, a similar pattern emerges in that smaller deletion sizes dominate the distribution. It is important to note that whereas a few of the deletions seen at the megabase level of resolution would theoretically be detectable by microscopy (e.g., banding analysis) virtually all of the deletions seen at the kilobase level would escape such detection. One of the two isolated clones that survived exposure to 56Fe ions resulted in a homogeneous reciprocal translocation between chromosomes 4 and 11. aCGH analysis showed that log2 ratios were significantly lower than zero within a section of chromosome 4, indicating the presence of a deletion. The deletion in question spanned several megabases of the long arm of that chromosome. Because it resides near the breakpoint of the translocation, it is likely that this deletion is involved in the rearrangement itself as a “codeletion”. This would imply that the 4:11 translocation is not reciprocal, but is actually part of a complex exchange not otherwise visible by cytogenetic techniques, in this case mFISH.

Bibliography Type:	Description: (Last Updated: 01/14/2026)
Abstracts for Journals and Proceedings	Loucas BD, Eberle RL, Yu Y, Cornforth MN. "Chromosome Aberrations Formed in Response to Changes in Proton Energy and Dose Rate." 21st Annual NASA Space Radiation Investigators; Workshop, Port Jefferson, NY, May 16-19th, 2010. Abstracts, 21st Annual NASA Space Radiation Investigators; Workshop, Port Jefferson, NY, May 16-19th, 2010. , May-2010
Abstracts for Journals and Proceedings	Ponomarev AL, Cornforth MN, Loucas B, Cucinotta F. "The distribution of small rings and associated deletions in human chromosomes as predicted by a generalized model of radiation-induced aberration formation." 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. Abstracts, 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=083eb449-219e-4aaf-aa64-2cf6533e153b&cKey=ce5da7d4-1791-4387-b2b8-eab66e232122&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Abstracts for Journals and Proceedings	Loucas BD, Eberle RL, Cornforth MN. "Irradiation of immortalized human fibroblasts with high energy protons at both low and high dose rates." 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. Abstracts, 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=8e70ff83-d8c8-427d-89cb-88fa6359360&cKey=26e3cae0-6093-4a96-9148-0928bd5201fc&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Peer-reviewed Journals	Bailey SM, Williams ES, Cornforth MN, Goodwin EH. "Chromosome Orientation fluorescence in situ hybridization or strand-specific FISH." Methods Mol Biol. 2010;659:173-83. PMID: 20809311 , Oct-2010